

2021-Ph-H1-Q18

Section: Particles and Waves

Topic: Spectra

An absorption spectrum is observed from starlight. What causes the dark lines?

Solution:

Cool gases **absorb specific wavelengths** of light as photons are absorbed by atoms.

Answer: C

Guidance for Students:

In an absorption spectrum, atoms absorb photons at exact energies, removing those wavelengths from the spectrum.

Revision Tip:

Emission = coloured lines on black

Absorption = black lines on colour

Both show electron transitions